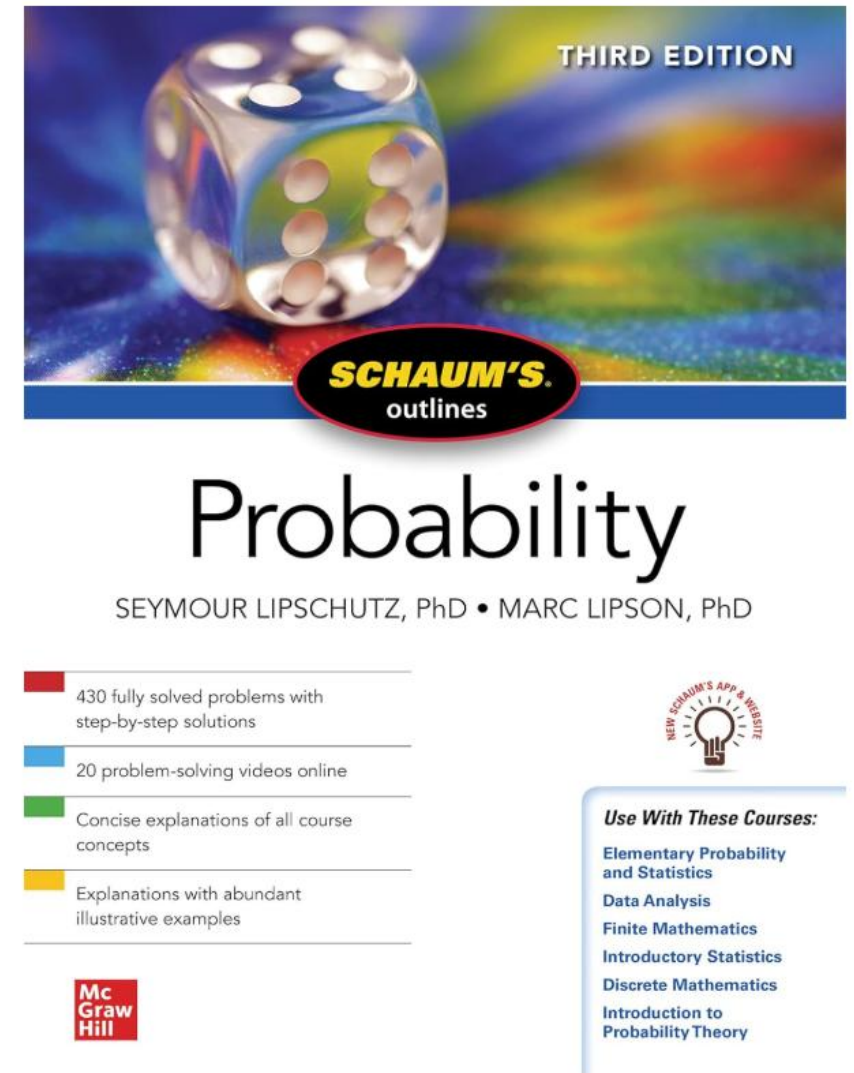


Probability and Statistics

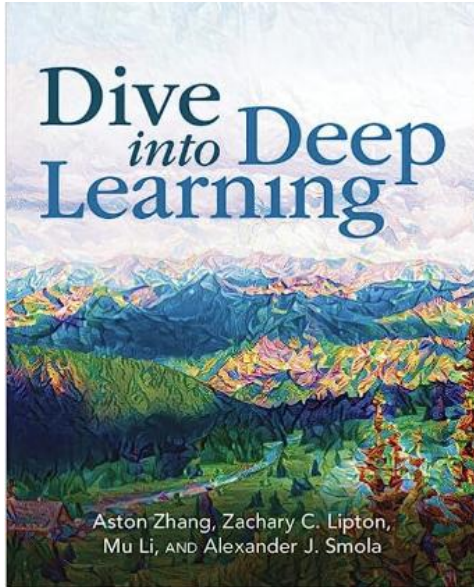
Avi Mohan

References

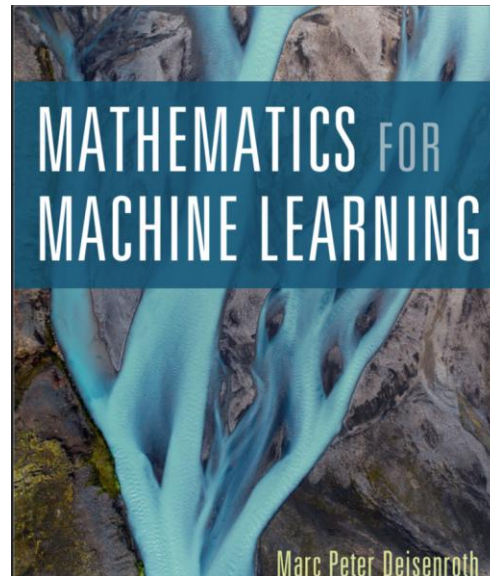
- Seymour Lipschutz, “***Probability,***” ***Schaum’s Outline of Probability***, 3rd Edition.
- Solved problems
- I have reserved this book in the SEL library.
 - Please email either **Tracy Stone** (tstone@bu.edu), or **Sara Goodwin** (circulation@bu.libanswers.com) if you aren’t able to access the books.



Other References



- “Dive into Deep Learning,”
Sec. 2.6



- “Mathematics for Machine
Learning,” Chapter 6

Why Probability?

- Quantifying uncertainty.
 - Phenomena that involve chance or randomness
- ML (more importantly DL) are inherently probabilistic
 - Unsupervised learning: tSNE
 - *Quantifying estimator bias* and variance (tradeoff)
 - NN classifiers produce likelihoods
 - VAEs impose probability distributions over latent variables
 - Nash Equilibria of 2-player zero-sum games (GANs)

Today's class

- Probability basics
 - Sample space and events
- Conditional probability and Independence
- Random Variables
 - PMFs, PDFs
 - CDFs
- Expectation
 - Variance
 - Covariance and Correlation
- Estimators and Estimator “Bias”

Probability basics

- **Random experiment**: one whose outcome cannot be exactly predicted *a priori*
 - Rolling a die
- A. **Sample set**: Set of all possible outcomes (S)
- B. **Event**: an outcome/set of outcomes
 - Set of all events: **event set** ($\mathcal{C}$)
 - Events satisfy some rules
- C. **Probability** $\mathbb{P}$ is calculated for events
 - $\mathbb{P}[A]$ is the “probability of event A ”
 - A must be an event.

Examples

- $S = \{1, 2, \dots, 6\}$
- Events:
 - E_1 : An even number appears
 - E_2 : A number ≤ 3 appears
 - $\mathcal{C}$ contains all such sets
- $\mathbb{P}: E \rightarrow [0, 1]$
 - $\mathbb{P}[i] = \frac{1}{6}$ for every $i \in \{1, 2, \dots, 6\}$
 - $\mathbb{P}[E_1]$
 - $\mathbb{P}[E_2]$



Probability basics

- $\mathbb{P}[A^C] = 1 - \mathbb{P}[A]$
- $\mathbb{P}[S] = 1$
- $\Rightarrow \mathbb{P}[\emptyset] = 0$
- $\mathbb{P}[A \cup B] = \mathbb{P}[A] + \mathbb{P}[B] - \mathbb{P}[A \cap B]$

Conditional Probability and Independence

- Sometimes, the occurrence of an event can influence the chances of another event's occurrence
- Auto insurance:
 - A : accident occurs
 - E : driver is male and under 25
 - $\mathbb{P}[A] < \mathbb{P}[A|E]$
- Conditional Probability is concerned with *quantifying* such dependence.

Conditional Probability

- The conditional probability of event A given event E

$$\mathbb{P}[A|E] = \frac{\mathbb{P}[A \cap E]}{\mathbb{P}[E]}$$

- (⚠ $\mathbb{P}[E] > 0$)
- Example: rolling a pair of dice
 - What is S ?
 - E = sum is 6
 - A = 2 appears on at least one die



Conditional probability

- $\mathbb{P}[A \cap B] = \mathbb{P}[B]\mathbb{P}[A|B]$
- $\mathbb{P}[A \cap B \cap C] = \mathbb{P}[A] \times \mathbb{P}[B|A] \times \mathbb{P}[C|A \cap B]$
 - The probability that events A , B , and C occur is the product of
 - The probability that A occurs,
 - The probability that B occurs given A has occurred
 - The probability that C occurs given A and B have occurred.
- Example: A lot contains 12 items & 4 are defective
 - Pick 3 items at random without replacement.
 - ? Probability that all three are nondefective

Independent Events

- Events A and B are independent if the occurrence of one doesn't influence the occurrence of the other

$$\mathbb{P}[A|B] = \mathbb{P}[A]$$

- $\mathbb{P}[A \cap B] = \mathbb{P}[A] \times \mathbb{P}[B]$
- Independence is a *symmetric* relationship
- ⚠ Pairwise independence $\neq$ Independence!

Independent Events

- 3 independent coins are tossed.
 - What is S ?
- Three events
 - $A = \{HHH, HHT, HTH, HTT\}$
 - $B = \{HHH, HHT, THH, THT\}$
 - $C = \{HHT, THH\}$

Random Variables

- Neither “random” nor a “variable!”
- Given a random experiment with sample space S , a random variable X is a function from S to $\mathbb{R}$

$$X: S \rightarrow \mathbb{R}$$

- Needs to satisfy some properties
 - We will not concern ourselves with them.
- Example: Rolling a pair of dice
 - $X(a, b) = \max(a, b)$



Probability Mass Functions

- X : **finite** random variable

$$X: S \rightarrow \{x_1, x_2, \dots, x_n\}$$

- PMF $f_X(x_k) = \mathbb{P}[X = x_k]$

- $f_X(x_k) = \mathbb{P}[\{s \in S: X(s) = x_k\}]$

- “collect all the sample point s that the random variable X maps to x_k and find their probability.”

PMF: example

- Rolling a pair of dice
 - $X(a, b) = \max(a, b)$
 - What are S and R_X ?
- Computing $f_X(x)$ for $x \in \{1, 2, \dots, 6\}$



Cumulative Distribution Function

- Cumulative Distribution Function of X
 - Input: some real number a
 - Output: the probability that $X \leq a$
- Denoted by F_X

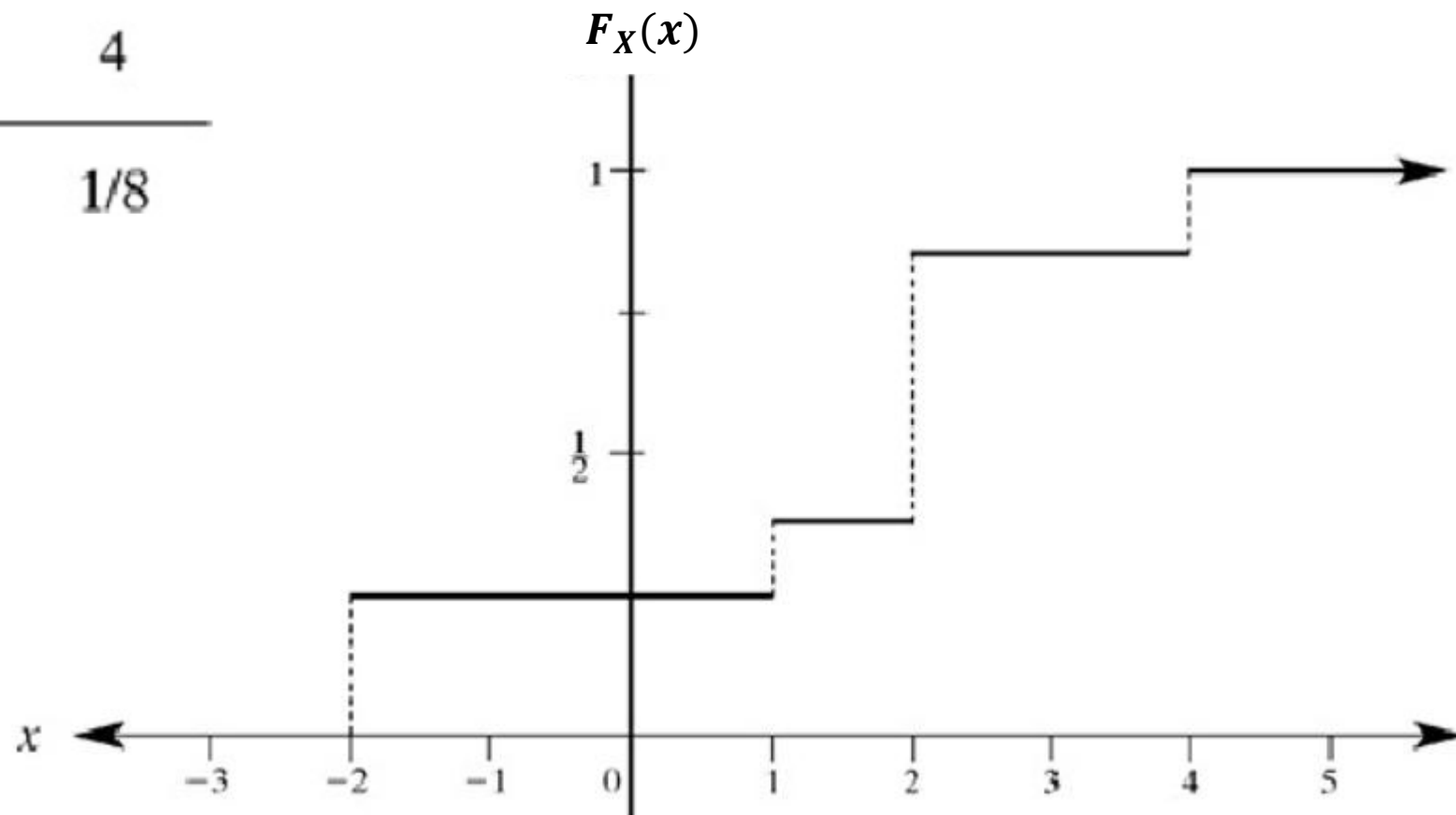
$$F_X(a) = \mathbb{P}[X \leq a]$$

- Example: X takes values $\{x_1, x_2, \dots, x_n\}$

$$F_X(a) = \sum_{x_i \leq a} f_X(x_i)$$

CDF example

x	-2	1	2	4
$f(x)$	$1/4$	$1/8$	$1/2$	$1/8$



CDF example

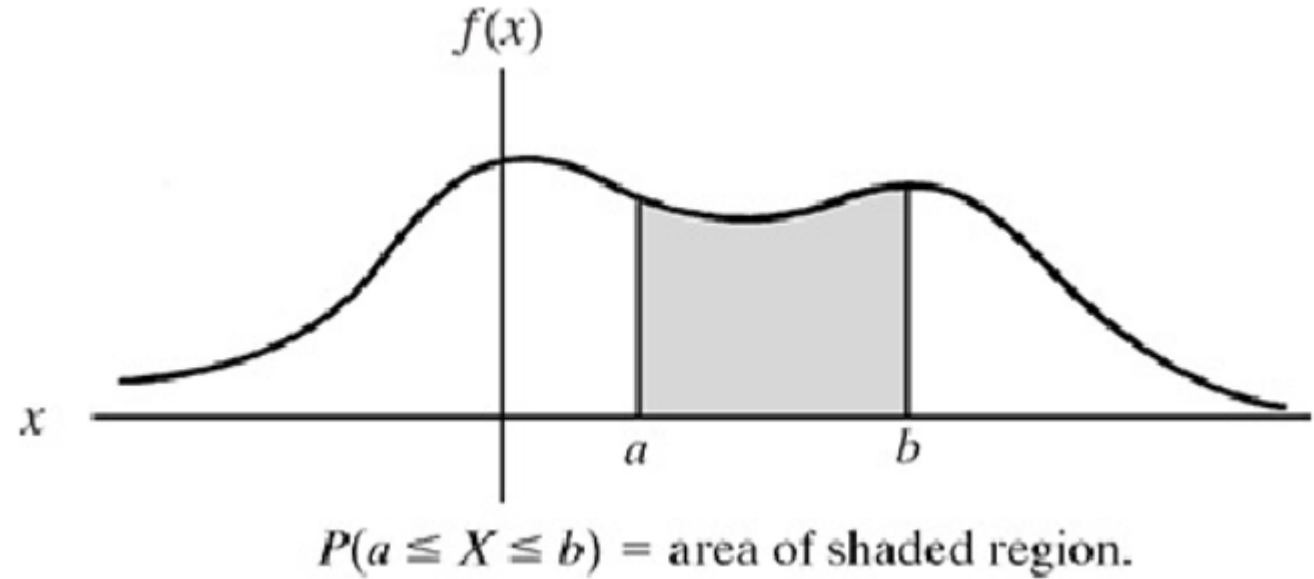
- Roll a pair of dice. Say, the outcome is (a, b)
 - Random variable $Y = a + b$
- What is the CDF F_Y ?

CDF properties

- F_X is nondecreasing
- $\lim_{a \rightarrow -\infty} F_X(a) = 0$
- $\lim_{a \rightarrow \infty} F_X(a) = 1$
- (outside course scope) F_X is right-continuous

Continuous RVs

- The range of values X can take is continuous
- Now, the PDF $f_X(a)$ at a single point doesn't make sense.



$$\mathbb{P}[a \leq X \leq b] = \int_a^b f_X(x) dx$$

PDFs commonly used in ML

- **Bernoulli distribution**

- Two outcomes: “success,” and “failure”
- X maps “success” to 1, and “failure” to 0
- $\mathbb{P}[\text{success}] = \mathbb{P}[X = 1] = p$

- PDF $f_X(x) = \begin{cases} p, & x = 1 \\ 1 - p, & x = 0 \end{cases}$

- CDF F_X

Binomial distribution

- Number of “successes” in n IID Bernoulli trials
 - *Independent and identically distributed*
- PDF f_X also denoted by $B(n, k)$
 - Probability of k successes in n Bernoulli trials

$$B(n, k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

- What is $\sum_{k=1}^n B(n, k)$?

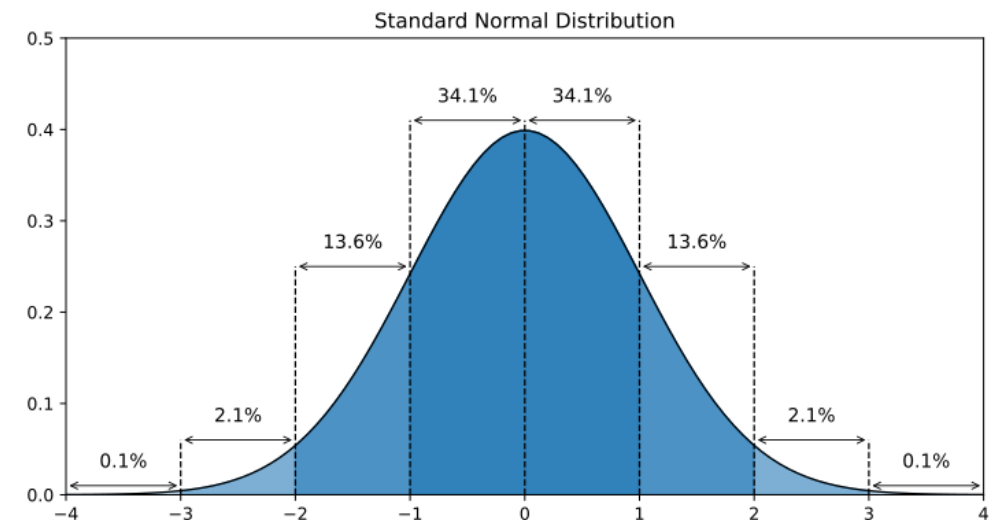
Normal Distribution

- Many data sets and observations of natural phenomena tended to follow a “bell” curve
 - Doesn't mean the others are abnormal in any way.

- X is said to be “normally” distributed if

$$f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$$

- Aka the *Gaussian Distribution*



Uniform Distribution

- Discrete

- Range = $\{x_1, \dots, x_n\}$

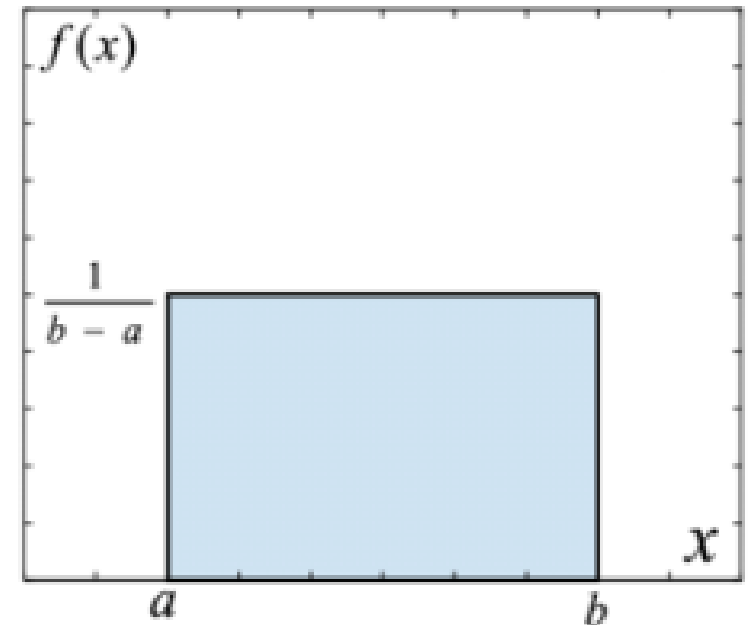
$$f_X(x_i) = \frac{1}{n}$$

- Continuous

- Range = $[a, b]$

$$f_X(x) = \frac{1}{b - a}$$

- What do the CDFs look like?



Geometric Distribution

- Infinite series of IID Bernoulli trials
 - Probability of k consecutive successes before a failure
- $f_X(k) = \mathbb{P}[X = k] = p^k(1 - p)$
- Also denoted by $Geo(p)$
- What is the CDF?

Statistics: mean, variance, and covariance

- Suppose X takes values in $\{x_1, \dots, x_n\}$
- The expected value of X

$$\mathbb{E}X = \sum_{i=1}^n x_i \mathbb{P}[X = x_i]$$

$$= \sum_{i=1}^n x_i f_X(x_i)$$

- Continuous RVs

Variance and Standard Deviation

- The variance of X is given by

$$\begin{aligned}\sigma^2 &= \mathbb{E}(X - \mathbb{E}X)^2 \\ &= \sum_{i=1}^n (x_i - \mathbb{E}X)^2\end{aligned}$$

- Standard deviation $\sigma = \sqrt{\mathbb{E}(X - \mathbb{E}X)^2}$

- $\mathcal{N}(\mu, \sigma^2)$

Multiple RVs and Covariance

- Suppose X and Y are RVs, and
 - $X \in \{x_1, \dots, x_n\}$ and
 - $Y \in \{y_1, \dots, y_m\}$
- $f_{XY}(x_i, y_j) = \mathbb{P}[X = x_i, Y = y_j]$
 - *Joint probability mass function*
 - $f_{XY}(x_i, y_j) \geq 0$

$$\sum_i \sum_j f_{XY}(x_i, y_j) = 1$$

Covariance

- We are interested in the degree to which two RVs move ($\uparrow, \downarrow$) together.
- Quantified by **covariance** and the **correlation coefficient**.

$$\begin{aligned} cov(X, Y) &= \sum_i \sum_j (x_i - \mathbb{E}X)(y_j - \mathbb{E}Y) f_{XY}(x_i, y_j) \\ &= \mathbb{E}(X - \mathbb{E}X)(Y - \mathbb{E}Y) \end{aligned}$$

Correlation Coefficient

$$\text{cov}(X, Y) = \mathbb{E}(X - \mathbb{E}X)(Y - \mathbb{E}Y)$$

The correlation coefficient $\rho(X, Y) = \frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y}$

- $-1 \leq \rho(X, Y) \leq 1$
- What are $\rho(X, X)$ and $\rho(X, -X)$?